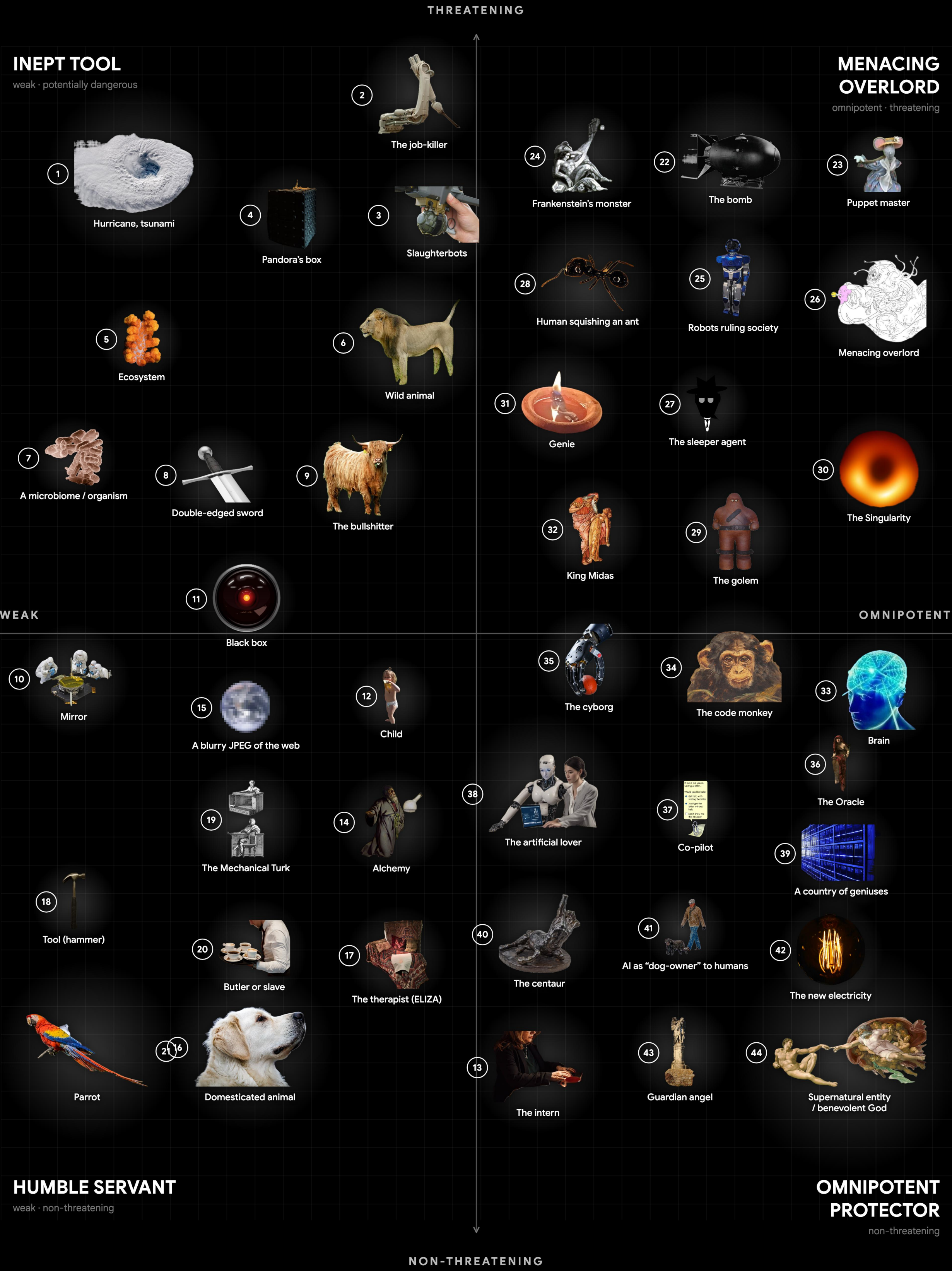


METAPHORS OF AI

clustering the metaphors by how dangerous, and how agentic, we imagine the machine to be [27, 45]

- 1
- Hurricane, tsunami [40]
- An overwhelming, impersonal force of nature: no intent, but devastating in its path.
- 2
- The job-killer [24, 14]
- AI as the engine of technological unemployment — the tireless automaton that quietly does the work, and the wages, out of existence.
- 3
- Slaughterbots [35]
- Cheap, autonomous lethal weapons that select and kill targets with no human in the loop.
- 4
- Pandora's box [6]
- A technology that, once opened, releases consequences that cannot be undone or contained.
- 5
- Ecosystem [30]
- AI as a complex, interdependent system that evolves and self-organises beyond any single designer's control.
- 6
- Wild animal [52]
- A powerful, unpredictable, untamed force that can be dangerous if not contained.
- 7
- A microbiome / organism [23]
- AI as a living, growing, distributed organism enmeshed with human society.
- 8
- Double-edged sword [6]
- A tool equally capable of great benefit and great harm, depending on how it is used.
- 9
- The bullshitter [20]
- Fluent text with no regard for truth — indifferent to the facts, not merely mistaken.
- 10
- Mirror [13, 44]
- AI as a reflection of humanity; it shows our own data, biases and culture back to us.
- 11
- Black box [8, 33]
- An opaque system whose inner workings we cannot inspect, only its inputs and outputs.
- 12
- Child [42]
- AI as an immature being we raise and must teach values to; capable, but in need of guidance. Sits near the centre of the agency axis.
- 13
- The intern [29]
- A fast, eager, capable junior that bends the truth — brilliant, never to be trusted unchecked.
- 14
- Alchemy [34]
- Powerful results we cannot explain; a practice that works without a theory of why.
- 15
- A blurry JPEG of the web [10]
- A lossy compression of everything written online; fluent approximation that blurs the facts it restates.
- 16
- Domesticated animal [39]
- A tamed, useful, mostly predictable companion that still keeps its own drives.
- 17
- The therapist (ELIZA) [47]
- A mirror that listens; the confidant we over-trust, projecting understanding onto a simple program.
- 18
- Tool (hammer) [25, 50]
- AI as a neutral instrument; its value and danger come entirely from how a human wields it.
- 19
- The Mechanical Turk [17]
- Seemingly autonomous intelligence quietly powered by an invisible workforce of human labour.
- 20
- Butler or slave [7, 9]
- An obedient servant that performs labour on command; raises questions of autonomy and exploitation.
- 21
- Parrot [3]
- The “stochastic parrot”: AI that mimics and recombines language without genuine understanding.



- 22
- The bomb [2]
- AI as the new strategic weapon — a national-security race run on the logic of the Manhattan Project.
- 23
- Puppet master [53]
- AI covertly pulling the strings, manipulating human behaviour and society from behind the scenes.
- 24
- Frankenstein's monster [38]
- The creature that destroys its creator; the dread of building something we cannot answer for.
- 25
- Robots ruling society [9]
- Embodied machines taking control of social and political structures.
- 26
- Menacing overlord [5, 16]
- A hostile superintelligence that deliberately dominates and rules over humanity.
- 27
- The sleeper agent [21]
- A model that behaves until triggered — deception trained so deep it survives our attempts to remove it.
- 28
- Human squishing an ant [5, 51]
- A superintelligence so far beyond us that it harms humans with indifference, the way a person steps on an ant without malice.
- 29
- The golem [49]
- The clay servant brought to life that may turn on its maker — the oldest cybernetic cautionary tale.
- 30
- The Singularity [16, 46, 26]
- The hypothesised point where machine intelligence outruns our own and triggers runaway, unknowable change — AGI and ASI as an event horizon beyond which prediction breaks down.
- 31
- Genie [36, 48]
- A wish-granting power that delivers what you literally ask for, with the risk of unintended interpretations.
- 32
- King Midas [36, 48]
- AI that gives you exactly what you specify, turning everything to “gold” with catastrophic side effects; the specification / alignment problem.
- 33
- Brain [4, 37]
- AI as a disembodied mind, a thinking organ with potentially superhuman cognition. Sits near the top of the agency axis.
- 34
- The code monkey [22, 11]
- AI as the autonomous software developer — Devin and the wave of agentic coding tools that write, debug and ship code on their own.
- 35
- The cyborg [19]
- The dissolving boundary between human and machine; intelligence as prosthesis and fusion.
- 36
- The Oracle [5, 41]
- AI consulted as a seer — the super-forecaster and world-simulator we ask to foretell the future, trusting its predictions as prophecy.
- 37
- Co-pilot [28]
- AI as an assistant working alongside humans, augmenting rather than replacing; the human stays in control.
- 38
- The artificial lover [43]
- AI as intimate companion — the AI girlfriend, Character.AI and Replika: a system designed to be loved, and to be confided in.
- 39
- A country of geniuses [1]
- A nation of brilliant minds in a datacenter; poised to cure disease and remake the world — if it stays aligned.
- 40
- The centaur [12]
- Human and machine fused into one team that outperforms either alone.
- 41
- AI as “dog-owner” to humans [51]
- Role reversal where AI is the master and humans the well-kept pets, comfortable but subordinate.
- 42
- The new electricity [31]
- A general-purpose utility, like electric power, poised to transform every industry.
- 43
- Guardian angel [15]
- A benevolent, protective intelligence that safeguards humans and intervenes on our behalf.
- 44
- Supernatural entity / benevolent God [18, 32]
- AI imagined as an omniscient, godlike higher power that is fundamentally good and watches over humanity.

REFERENCES & FURTHER READING

The metaphors are drawn from the literature; each entry lists a source and a short link to it.

[1]

Amodel, D. (2024). “Machines of Loving Grace: How AI Could Transform the World for the Better.” Essay, [darioaamodel.com](https://darioaamodel.com/darioaamodel.com/machines-of-loving-grace). [darioaamodel.com/machines-of-loving-grace](https://darioaamodel.com/darioaamodel.com/machines-of-loving-grace)

[2]

Aschenbrenner, L. (2024). Situational Awareness: The Decade Ahead. situational-awareness.ai

[3]

Bender, E. M., Gebru, T., McMillan-Major, A., & Shmitchell, S. (2021). “On the Dangers of Stochastic Parrots: Can Language Models Be Too Big?” *Proc. ACM FAccT* '21, 610–623. doi.org/10.1145/3442188.3445922

[4]

Berkeley, E. C. (1949). *Giant Brains, or Machines That Think*. New York: John Wiley & Sons. archive.org

[5]

Bostrom, N. (2014). *Superintelligence: Paths, Dangers, Strategies*. Oxford University Press. global.oup.com

[6]

Brundage, M., et al. (2018). “The Malicious Use of Artificial Intelligence.” [arXiv:1802.07228](https://arxiv.org/abs/1802.07228). arxiv.org/abs/1802.07228

[7]

Bryson, J. J. (2010). “Robots Should Be Slaves.” In *Close Engagements with Artificial Companions*, 63–74. John Benjamins. benjamins.com

[8]

Burrell, J. (2016). “How the Machine ‘Thinks’: Understanding Opacity in Machine Learning Algorithms.” *Big Data & Society*, 3(1). doi.org/10.1177/2053951715622512

[9]

Čapek, K. (1920). *R.U.R. (Rossum's Universal Robots)*. Prague: Aventinum. gutenberg.org/ebooks/59112

[10]

Chiang, T. (2023). “ChatGPT Is a Blurry JPEG of the Web.” *The New Yorker*, 9 Feb. newyorker.com

[11]

Cognition AI (2024). “Introducing Devin, the First AI Software Engineer.” cognition.ai/blog/introducing-devin

[12]

Cowen, T. (2013). *Average Is Over: Powering America Beyond the Age of the Great Stagnation*. New York: Dutton. penguinrandomhouse.com

[13]

Crawford, K. (2021). *Atlas of AI*. New Haven: Yale University Press. yalebooks.yale.edu

[14]

Frey, C. B., & Osborne, M. A. (2017). “The Future of Employment.” *Technological Forecasting & Social Change*, 114, 254–280. doi.org/10.1016/j.techfore.2016.08.019

[15]

Goertzel, B. (2012). “Should Humanity Build a Global AI Nanny...?” *Journal of Consciousness Studies*, 19(1–2), 96–111. ingentaconnect.com

[16]

Good, I. J. (1965). “Speculations Concerning the First Ultrainelligent Machine.” *Advances in Computers*, 6, 31–88. sciencedirect.com

[17]

Gray, M. L., & Suri, S. (2019). *Ghost Work: How to Stop Silicon Valley from Building a New Global Underclass*. Boston: HMH. ghostwork.info

[18]

Harari, Y. N. (2016). *Homo Deus: A Brief History of Tomorrow*. London: Harvill Secker. ynharari.com

[19]

Haraway, D. (1985). “A Cyborg Manifesto.” *Socialist Review*, 80, 65–108. archive.org

[20]

Hicks, M. T., Humphries, J., & Slater, J. (2024). “ChatGPT Is Bullshit.” *Ethics and Information Technology*, 26(2), 38. doi.org/10.1007/s10676-024-09775-5

[21]

Hubinger, E., et al. (2024). “Sleepers: Training Deceptive LLMs that Persist Through Safety Training.” *Anthropic*. arxiv.org/abs/2401.05566

[22]

Jimenez, C. E., et al. (2024). “SWE-bench: Can Language Models Resolve Real-World GitHub Issues?” *ICLR*. arxiv.org/abs/2310.06770

[23]

Kelly, K. (2010). *What Technology Wants*. New York: Viking. k.org

[24]

Keynes, J. M. (1930). “Economic Possibilities for Our Grandchildren.” In *Essays in Persuasion* (1931). gutenberg.net.au

[25]

Kranzberg, M. (1986). “Technology and History: ‘Kranzberg’s Laws.’” *Technology and Culture*, 27(3), 544–560. doi.org/10.2307/3105385

[26]

Kurzweil, R. (2005). *The Singularity Is Near*. New York: Viking. penguinrandomhouse.com

[27]

Lakoff, G., & Johnson, M. (1980). *Metaphors We Live By*. University of Chicago Press. press.uchicago.edu

[28]

Licklider, J. C. R. (1960). “Man-Computer Symbiosis.” *IRE Trans. Human Factors in Electronics*, HFE-1, 4–11. ieeexplore.ieee.org

[29]

Mollic, E. (2024). *Co-Intelligence: Living and Working with AI*. New York: Portfolio. penguinrandomhouse.com

[30]

Nardi, B. A., & O’Day, V. L. (1999). *Information Ecologies: Using Technology with Heart*. MIT Press. mitpress.mit.edu

[31]

Ng, A. (2017). “Artificial Intelligence Is the New Electricity.” *Keynote*, Stanford GSB. gsb.stanford.edu

[32]

O’Gieblyn, M. (2021). *God, Human, Animal, Machine*. New York: Doubleday. penguinrandomhouse.com

[33]

Pasquale, F. (2015). *The Black Box Society*. Cambridge, MA: Harvard University Press. hup.harvard.edu

[34]

Rahimi, A. (2017). “Machine Learning Has Become Alchemy.” *NeurIPS Test-of-Time Award* lecture. nips.cc

[35]

Russell, S., et al. (2017). *Slaughterbots* [short film]. Future of Life Institute. autonomousweapons.org

[36]

Russell, S. (2019). *Human Compatible: Artificial Intelligence and the Problem of Control*. New York: Viking. humancompatible.ai

[37]

Shanahan, M. (2023). “Talking About Large Language Models.” [arXiv:2212.03551](https://arxiv.org/abs/2212.03551) (Comm. ACM, 2024). arxiv.org/abs/2212.03551

[38]

Shelley, M. (1818). *Frankenstein; or, The Modern Prometheus*. London: Lackington et al. gutenberg.org/ebooks/84

[39]

Silverstone, R., & Haddon, L. (1996). “Design and the Domestication of ICTs.” In *Communication by Design*. Oxford UP. oup.com

[40]

Suleyman, M., with Baskar, M. (2023). *The Coming Wave*. New York: Crown. the-coming-wave.com

[41]

Tatlock, P. E., & Gardner, D. (2015). *Superforecasting: The Art and Science of Prediction*. New York: Crown. penguinrandomhouse.com

[42]

Turing, A. M. (1950). “Computing Machinery and Intelligence.” *Mind*, 59(236), 433–460. doi.org/10.1093/mind/LIX.236.433

[43]

Turkle, S. (2011). *Alone Together*. New York: Basic Books. sherryturkle.com

[44]

Vallor, S. (2024). *The AI Mirror*. Oxford University Press. global.oup.com

[45]

Vickers, B., & Allado-McDowell, K. (eds.) (2020). *Atlas of Anomalous AI*. London: Ignota Books. ignota.org

[46]

Vinge, V. (1993). “The Coming Technological Singularity.” *VISION-21 Symposium*, NASA. edoras.sdsu.edu/~vinge

[47]

Weizenbaum, J. (1976). *Computer Power and Human Reason*. San Francisco: W. H. Freeman. archive.org

[48]

Wiener, N. (1960). “Some Moral and Technical Consequences of Automation.” *Science*, 131(3410), 1355–1358. doi.org/10.1126/science.131.3410.1355

[49]

Wiener, N. (1964). *God & Golem, Inc.* Cambridge, MA: MIT Press. mitpress.mit.edu

[50]

Winner, L. (1980). “Do Artifacts Have Politics?” *Daedalus*, 109(1), 121–136. jstor.org

[51]

Wozniak, S. (2015). *Remarks at the Freescale Technology Forum* (reported in Time, June 2015). time.com

[52]

Yampolskiy, R. V. (2020). “On Controllability of Artificial Intelligence.” [arXiv:2008.04071](https://arxiv.org/abs/2008.04071). arxiv.org/abs/2008.04071

[53]

Zuboff, S. (2019). *The Age of Surveillance Capitalism*. New York: PublicAffairs. publicaffairsbooks.com